Theorems

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1 Chapter 1: Convergence of Sequences

1.1 1.1 Metric Spaces

- Definition 1.1.1: Metric [Chance: 80%]
  A mapping $d_\Omega : \Omega \times \Omega \to [0, \infty)$ is a metric if it satisfies:
  - (M1) Symmetry: $d_\Omega(x, y) = d_\Omega(y, x)$
  - (M2) Triangle Inequality: $d_\Omega(x, z) \leq d_\Omega(x, y) + d_\Omega(y, z)$
  - (M3) Definiteness: $d_\Omega(x, y) = 0 \iff x = y$
- Definition 1.1.2: Convergence [Chance: 80%]
  A sequence (a_n) converges to a if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N, d_\Omega(a_n, a) \leq \epsilon$.
- Example 1.1.4: Basic Metric Spaces [Chance: 50%]
  - Real line: $d(x, y) = |x - y|$.
  - Plane with taxicab distance: $d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$.
- Plane with maximum distance: $d_{max}(x, y) = max\{|x_1 - y_1|, |x_2 - y_2|\}$.
  - Discrete metric: $d(x, y) = 0$ if $x = y$, else 1.
- Theorem 1.1.6: Uniqueness of Limit [Chance: 50%]
  If $a_n \to a$ and $a_n \to a'$, then $a = a'$.
- Theorem 1.1.7: Non-negativity [Chance: 50%]
  If $(a_n) \subseteq \mathbb{R}$ and $a_n \to a$, then $\forall n : a_n \geq 0 \Rightarrow a \geq 0$.

1.2 1.2 Completeness

- Definition 1.2.1: Cauchy Sequence [Chance: 70%]
  A sequence is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n, m \geq N, d_\Omega(a_n, a_m) \leq \epsilon$.
- Lemma 1.2.3: Convergence implies Cauchy [Chance: 60%]
  Every convergent sequence is a Cauchy sequence.
- Theorem 1.2.12: Bolzano-Weierstraß [Chance: 95%]
  Every bounded sequence in $\mathbb{R}$ contains a convergent subsequence.
- Theorem 1.2.14: Monotone Convergence [Chance: 80%]
  If $(a_n) \subseteq \mathbb{R}$ is monotonically increasing and bounded from above, it is convergent.
- Theorem 1.2.16: Completeness of C [Chance: 60%]
  The space of complex numbers $\mathbb{C}$ is complete.

1.3 1.3 Compactness

- Definition 1.3.2: Compactness [Chance: 0%]

  A set K is compact if every sequence in K has a convergent subsequence whose limit lies in K.

- Theorem 1.3.7: Heine-Borel (in R) [Chance: 0%]

  A set $K \subseteq \mathbb{R}$ is compact if and only if it is closed and bounded.

2 Chapter 2: Continuity and Special Functions

2.1 2.1 Series and the Exponential Function

- Theorem 2.1.4: Geometric Series [Chance: 90%]

  If $|x| < 1$, then $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$.

- Theorem 2.1.12: Convergence Criteria [Chance: 95%]

  - Comparison test: If $|a_n| \leq |b_n|$ and $\sum |b_n|$ converges, then $\sum |a_n|$ converges.

- Ratio test: If $|\frac{a_{n+1}}{a_n}| \leq \theta < 1$ for $n \geq N$, then $\sum |a_n|$ converges.

- Leibniz criterion: If a_n is non-negative, monotonically decreasing, and $a_n \to 0$, then the alternating series $\sum (-1)^n a_n$ converges.

- Theorem 2.1.23: Functional Equation of Exp [Chance: 75%]

  $exp(x + y) = exp(x)exp(y)$ for all $x, y \in \mathbb{C}$.

2.2 2.2 Continuity

- Definition 2.2.1: Continuity [Chance: 85%]

  f is continuous in x if for every sequence $x_n \to x$, we have $f(x_n) \to f(x)$.

- Theorem 2.2.7: Extreme Value Theorem [Chance: 80%]

  A continuous function f on a compact set K attains its maximum and minimum.

- Theorem 2.2.9: Intermediate Value Theorem [Chance: 80%]

  If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < 0 < f(b)$, there exists $\xi \in (a, b)$ such that $f(\xi) = 0$.

- Theorem 2.2.21: Logarithm [Chance: 85%]

  The inverse of $exp : \mathbb{R} \to (0, \infty)$ is the continuous, strictly increasing function $log : (0, \infty) \to \mathbb{R}$.

2.3 2.3 Trigonometric Functions

- Theorem 2.3.2: Series expansion [Chance: 95%]
  $cos(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k}}{(2k)!}$ and $sin(x) = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k+1}}{(2k+1)!}$.
- Theorem 2.3.9: Periodicity [Chance: 60%]
  $e^{i(x+2\pi k)} = e^{ix}$, $cos(x + 2\pi k) = cos(x)$, and $sin(x + 2\pi k) = sin(x)$ for $k \in \mathbb{Z}$.

3 Chapter 3: Differentiability

3.1 3.1 Definition and Examples

- Theorem 3.1.8: Linear Approximability [Chance: 85%]
  f is differentiable at x_0 iff $f(x) = f(x_0) + c(x - x_0) + \varphi(x)$ where $lim_{x\to x_0} \frac{\varphi(x)}{x-x_0} = 0$.

3.2 3.2 Differentiation Rules

- Summary of Rules [Chance: 90%]
  - Linearity: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.
- Product Rule: $(fg)' = f'g + fg'$.
  - Chain Rule: $(g \circ f)'(x) = g'(f(x))f'(x)$.
- Inverse Function: $(f^{-1})'(y_0) = \frac{1}{f'(f^{-1}(y_0))}$.

3.3 3.3 Local Extrema

- Theorem 3.3.2: Necessary Condition [Chance: 70%]
  If f has a local extremum at x_0, then $f'(x_0) = 0$.
- Theorem 3.3.10: Mean Value Theorem [Chance: 95%]
  If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $\xi \in (a, b)$ such that $\frac{f(b)-f(a)}{b-a} = f'(\xi)$.
- Theorem 3.3.14: Second Derivative Test [Chance: 75%]
  If $f'(x_0) = 0$ and $f''(x_0) > 0$, f has a strict local minimum. If $f''(x_0) < 0$, f has a strict local maximum.

3.4 3.4 Convex Functions

- Theorem 3.4.3: Second Derivative Test for Convexity [Chance: 40%]
  f is convex iff $f''(t) \geq 0$ for all t.
- Theorem 3.4.4: Global Minimizer [Chance: 40%]
  For convex, twice continuously differentiable functions, x^* is a global minimizer iff $f'(x^*) = 0$.

4 Chapter 4: Integration

4.1 4.1 Partitions and Darboux Sums

- Definition 4.1.8: Integrability [Chance: 70%]
  f is integrable if its lower integral equals its upper integral.
- Theorem 4.1.12: Continuity implies Integrability [Chance: 70%]
  Every continuous function on $[a, b]$ is integrable.

4.2 4.2 Fundamental Theorem of Calculus

- Theorem 4.2.3: FTC [Chance: 90%]
  If F is differentiable with continuous derivative F', then $\int_a^b F'(x)dx = F(b) - F(a)$.
- Integration Methods [Chance: 85%]
  - Integration by Parts: $\int f'g = fg - \int fg'$.
  - Substitution Rule: $\int_a^b f(u(x))u'(x)dx = \int_{u(a)}^{u(b)} f(y)dy$.

4.3 4.3 Taylor Expansion

- Theorem 4.3.1: Taylor Expansion [Chance: 95%]
  $f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_{n+1}(x)$ with $R_{n+1}(x) = \frac{1}{n!}\int_{x_0}^{x}(x - t)^n f^{(n+1)}(t)dt$.

4.4 4.4 Improper Integrals

- Theorem 4.4.4: Integral Test [Chance: 85%]
  For non-negative, monotonically decreasing f, the series $\sum f(n)$ converges if $\int_0^\infty f(x)dx$ exists.

4.5 4.5 Uniform Convergence

- Theorem 4.5.1: Uniform Convergence [Chance: 90%]

  If a sequence of continuous functions f_n converges uniformly to f, then f is continuous.